UID EXPERIMENT-10

AIM:

The aim is to create data visualizations such as pie charts and bar graphs to analyze semester result performance using JavaScript and Chart.js.

PROCEDURE:

Step 1: Create HTML File (index.html)

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Semester Result Analysis</title>

<style>

body {

    font-family: Arial;

    text-align: center;

    margin: 50px;

}


canvas {

    margin: 20px auto;

}

</style>
```

```
</head>
<body>
<h1>Semester Result Analysis System</h1>
<canvas id="pieChart" width="400" height="400"></canvas>
<canvas id="barChart" width="400" height="400"></canvas>
<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
<script src="script.js"></script>
</body>
</html>
```

Step 2: Create JavaScript File (script.js)

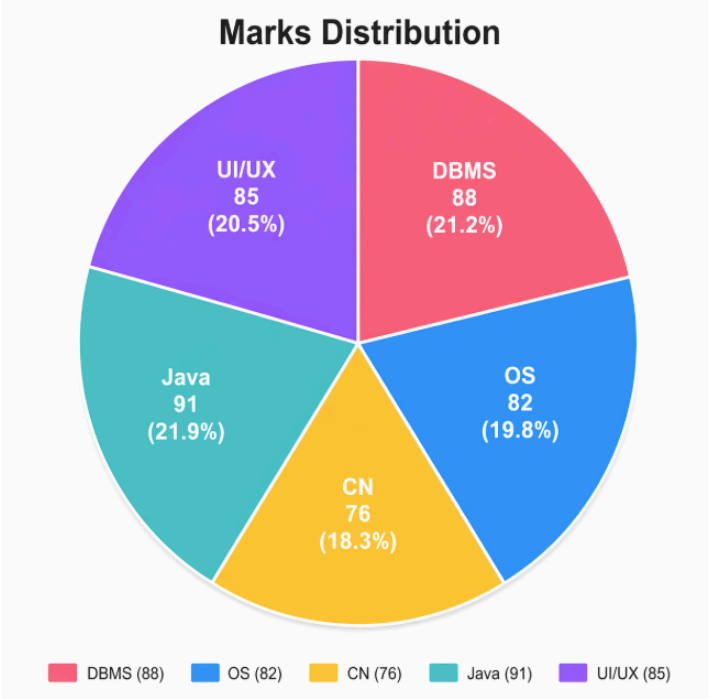
```
const semesterData = {
labels: ['DBMS', 'OS', 'CN', 'Java', 'UI/UX'],
datasets: [{
    label: 'Marks',
    data: [88, 82, 76, 91, 85],
    backgroundColor: [
        '#FF6384',
        '#36A2EB',
        '#FFCE56',
        '#4BC0C0',
        '#9966FF'
    ]
}]
};
const ctxPie = document.getElementById('pieChart').getContext('2d');
```

```
new Chart(ctxPie, {
type: 'pie',
data: semesterData,
options: {
    responsive: true,
    plugins: {
        title: {
            display: true,
            text: 'Marks Distribution'
        }
    }
}
});

// BAR CHART

const ctxBar = document.getElementById('barChart').getContext('2d');

new Chart(ctxBar, {
type: 'bar',
data: semesterData,
options: {
    responsive: true,
    plugins: {
        title: {
            display: true,
```

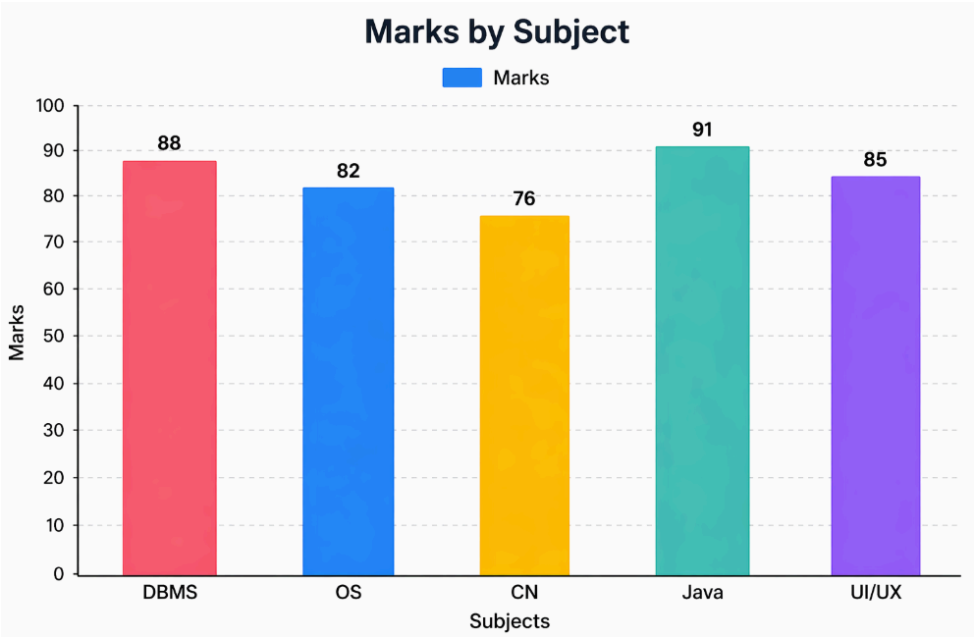
```
            text: 'Marks by Subject'
        }
    },
    scales: {
        y: {
            beginAtZero: true
        }
    }
}
});
```

OUTPUT:

Pie chart showing distribution of marks across subjects.



Bar chart comparing marks for each subject.



RESULT:

Thus, the semester result analysis visualization system was successfully created using JavaScript and Chart.js.